

RIBOSOME NETWORK

A decentralized RNA inverse-folding subnet: miners are synthetases that design RNA sequences for rotating structural targets; validators are chaperones that refold and score them. Commit-reveal, duplicate gates, diversity bonuses and delayed score reveals make every attack path earn zero — implemented, tested and simulated in this submission.

θ_{dup} 0.85 · w_{div} 0.1 · T_{rot} 16 · B 4 · δ_{tm} 0.35 · 58 tests
green

Ribosome Network Team

A synthetic ribosome on the Bittensor ecosystem

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1. Executive Summary

The Ribosome Network is a Bittensor subnet that produces validated RNA inverse-folding work: given a target RNA structure drawn from a rotating pool, miners called **synthetases** return candidate sequences, and validators called **chaperones** refold every candidate with an oracle and score its structural fidelity. The commodity is expensive to produce, cheap to verify with a folding oracle, and scored by objective structural metrics - exactly the shape of work a decentralized network should coordinate. The mechanism implemented for this hackathon turns the Ribosome Network preprint's interface stubs into a working subnet layer: a five-phase commit-reveal round, a duplicate gate, a diversity bonus, delayed score reveals, and deterministic target-pool rotation.

Everything in this proposal runs. The repository ships a pure-Python mechanism package with 58 passing tests, an adversarial offline simulation that pits honest miners against four attack strategies across 20 epochs, and the charts generated from those runs. The headline evidence is summarized below: Sybil clones fold as well as their victims yet earn exactly zero weight, leakers are rejected every time, and honest quality concentrates network reward.

98-99%

Sybil clones zeroed as duplicates

0.000

clone weight share across 20 epochs

100%

leaker commits rejected (40 of 40)

The miner technology comes from our companion work on stability-aware mRNA inverse design, which measured the Pareto frontier between structural fidelity and thermodynamic stability for a coupled genetic algorithm. That measured trade-off - with a sweet spot at low beta where most of the stability gain comes at essentially zero fidelity cost - becomes the scoring heart of our production miner. We are asking the judges for a Bittensor testnet slot and incubation to take a mechanism that already survives its own attack surface onto a live network with three validators and ten miners.

2. Problem and Digital Commodity

Designing an RNA sequence that folds into a prescribed structure is the core operation behind mRNA therapeutics, aptamers, ribozymes and synthetic switches. Modern mRNA medicine adds a second constraint: the designed sequence must not only fold correctly but also fold stably, because degradation rates track structural stability. The search is genuinely expensive - each candidate must be folded, and meaningful design explores a space that grows as 4^n - while verification is comparatively cheap once an oracle exists. Quality is not a matter of taste: base-pair F1, contact-map RMSD and TM-score are objective, reproducible measures. This asymmetry between costly production and cheap verification is precisely the contract a Bittensor subnet is designed to enforce.

The subnet therefore sells a digital commodity: scored, ranked (sequence to structure) design work for target structures chosen by the network. Consumers - mRNA therapeutic developers, vaccine design teams, synthetic biology laboratories - buy the network's cumulative inverse-folding capability through standard synapses rather than building in-house folding and search infrastructure. Each epoch the commodity is produced fresh against a moving target pool, so the network's capability cannot be bought once and replayed; it must be earned continuously.

Our companion preprint supplies the miner-side technology with measured numbers. A forward model trained on 1,589 SN-filter==1 OpenVaccine sequences (107 nt, 68 scored positions) reaches a group-aware 5-fold MCRMSE of 0.356 plus or minus 0.004, and a coupled genetic algorithm trading structure distance against predicted instability moves MFE from -33.1 to -51.9 kcal/mol across the measured beta sweep while quantifying exactly how much fidelity each unit of stability costs. Section 4 uses that measured frontier as the miner's ranking objective; the subnet mechanism is agnostic to which generator plugs in.

3. Subnet Architecture and Round Lifecycle

Two roles coordinate every epoch. Synthetases (miners) design sequences; chaperones (validators) refold, score and weight them. The design intentionally mirrors molecular biology: a ribosome synthesizes from a template and chaperone proteins verify the fold, so the network is a synthetic ribosome whose product is RNA design itself. Targets live in a pool of 32 active structures; every 16 epochs half the pool is swapped from a 54-structure reservoir (manifest in the repository, validated for bracket balance and oracle self-consistency).



Figure 1 - The five phases of one 15-minute epoch. Full expanded flowchart: docs/charts/chart_mechanism_flowchart.png

Phase	Clock	Actor	What is enforced
COMMIT	3 min	Synthetases	SHA-256 of the packed K=4 candidate payload, salted and bound to (epoch, target_id); nothing to steal, no post-hoc swaps
EVALUATE	6 min	Chaperones	Oracle refold of every candidate; TM + base-pair F1 scoring against the target; validity gate at 0.35; duplicate detection; composite sealed for B = 4 epochs
SET_WEIGHTS	2 min	Chaperones	Only scores whose B-epoch hold elapsed are consumed; weights normalized and set through Yuma consensus
REVEAL	2 min	Synthetases	Public (payload, salt); hash mismatch or missed reveal scores zero
ROTATE	2 min	Protocol	Half the target pool swapped every $T_{rot} = 16$ epochs; commits naming targets absent from the pool are rejected

Table 1 - Phase schedule, actors and enforcement

One mapping between the offline harness and the eventual testnet is worth stating plainly. In the pure-Python harness the private evaluation query and the public reveal are collapsed into a single ledger reveal inside the epoch; on Bittensor these are distinct phases, with the payload first passing point-to-point from miner to validator and becoming public only in REVEAL. The enforcement semantics - commit-first, hash-verified, delayed scores - are identical in both settings, which is why the simulation is meaningful evidence for the testnet design.

4. Miner Responsibilities and Tasks

Each epoch a synthetase is assigned one target structure, deterministically derived from its hotkey and the epoch so that every miner's history is auditable and every target keeps a stable population of miners - which is precisely the condition under which duplicate detection has teeth. The miner generates $K = 4$ candidate sequences for the target's dot-bracket, ranks them with local compute, packs the ranked candidates into one payload, and commits its hash. Generating several candidates is legitimate compute spending: only the best candidate earns the structural score, with diversity rewarded separately, so K is quality insurance rather than spam.

The generator family

Generators implement one interface - produce k valid sequences for a target - so the market of miners can range from simple to state-of-the-art without touching the mechanism. The bundled **StubGenerator** fills target-paired positions with complementary nucleotides and samples the rest: it is the honest floor. The bundled **GAGenerator** implements the coupled objective from our stability-aware design preprint: candidates are selected by TM-score of their oracle fold minus beta times a thermodynamic-instability proxy, with tournament selection, crossover and Lamarckian repair of target-pair positions. Production

miners upgrade the instability term to the trained XGBoost forward model (MCRMSE 0.356) and the structural term to full folding physics - the interface is unchanged.

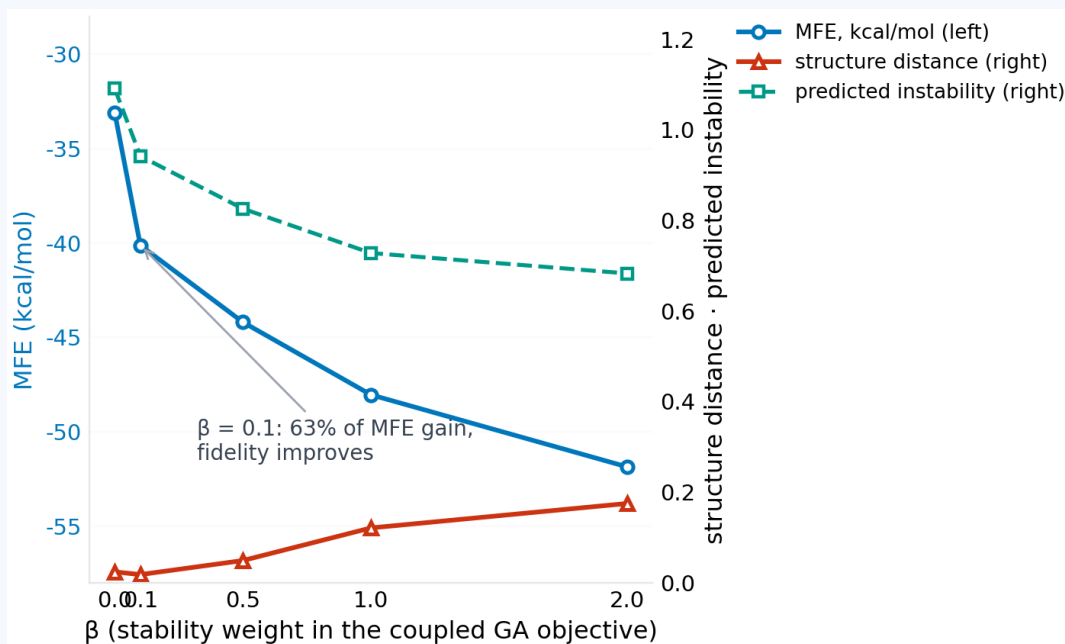


Figure 2 - The measured beta-Pareto frontier from the companion preprint (30 target structures; means over the beta sweep). The subnet's operating guidance: $\beta = 0.1-0.5$ where fidelity matters.

The frontier matters because it converts 'stable designs' from a vibe into a measured trade-off: at $\beta = 0.1$ the GA captures 63 percent of the achievable MFE improvement while slightly improving fold fidelity over a pure-structure search, while β of 1.0 and above erodes fidelity to the point where 13 to 19 positions disagree with the target. A miner that wants network reward must clear the 0.35 structural gate first; the coupled objective is how it buys stability with the fidelity it can spare.

5. Validator Responsibilities and Evaluation

The chaperone's evaluation pipeline is the subnet's source of truth, so it is deliberately mechanical. After the COMMIT phase closes, the validator walks every committed hotkey in arrival order: it rejects targets absent from the pool, drops unrevealed commitments, verifies the reveal's hash, refolds every revealed candidate with the oracle, scores the best candidate against the target, applies the validity gate, and only then lets survivors into duplicate detection and diversity accounting. Sealed scores are stored with a reveal epoch of $B = 4$; the weights staged in the same epoch consume only scores whose hold has elapsed.

The oracle chain

Oracle choice is a configuration, not a code path. The bundled **StubOracle** is a memoized Nussinov folder in pure Python/numpy - about ten milliseconds per fold at length 90 - which is what the simulation and the tests run on. The **ViennaRNAOracle** shells out to RNAfold for physics-grade secondary-structure folding

when the binary is present. The **RhoFoldOracle** slot awaits a RhoFold+ checkpoint in Phase 4, at which point the 2D base-pair F1 proxy upgrades to a 3D TM-score and contact-map RMSD without any change to the mechanism above it. Metrics already shipped: base-pair F1, contact-map RMSD, and the composite gate.

Validators are themselves scored by consensus: each validator's weights are compared against the quorum and deviations cost the deviator stake through Yuma. The quorum (5 validators) and trimmed mean (trim 1) parameters ship in the mechanism constants, and the harness records per-epoch sealed scores so validator disagreement is auditable from the logs.

6. Incentive and Reward Mechanism

The reward function is short enough to state completely. For miner i with best candidate folded to structure p against target t : the structural term is the base-pair F1 between p and t (upgraded to a true TM-score when the 3D oracle ships); the expression term is the fidelity proxy in 2D and the stability forward model in production; a candidate whose structural term falls below the 0.35 gate earns nothing. The composite is 0.7 times the structural term plus 0.3 times the expression term, a diversity bonus of 0.1 times the miner's mean pairwise 3-mer distance to the other accepted miners is added, and the result is max-normalized into the weight vector Yuma consumes.

$S = 0.7 \cdot \text{TM} + 0.3 \cdot \text{exp}$ validity-gated composite	$+ 0.1 \cdot \text{diversity}$ mean pairwise k-mer distance	$B = 4 \text{ epochs}$ score seal before weights consume it
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Each defense exists because a specific failure mode pays otherwise. Commit-first closes sequence theft: there is nothing to copy until the author is already credited. The duplicate gate closes Sybil farming: copying a winner's output is free to attempt and earns exactly zero, because the later commit by arrival order is the one zeroed. Delayed score reveals close oracle grinding: by the time a score is public the weights that consumed it are four epochs old, so probing the validator teaches nothing actionable. Pool rotation closes memorization: half the target pool is replaced every 16 epochs, so overfit strategies decay on a schedule. Rejected unknown target ids close leakage: a miner claiming to know the future pool is either leaking or faulty, and both cases score zero.

The emission flow follows standard Bittensor economics: the chain emits TAO to the subnet proportional to its registered stake and Yuma-normalized weights; validators take their configured cut; the remainder flows to miners in proportion to their normalized final scores. Because the diversity bonus re-normalizes only among accepted miners, the mechanism has no mechanism-level subsidy for failure - every zero is a zero.

7. Scoring Methodology and Anti-Gaming

The full scoring stack, in evaluation order, is tabulated below. Every parameter is a named constant in the mechanism package - there are no magic numbers buried in code paths - and every enforcement step writes a structured event to the run log, so any weight a validator sets can be replayed from the ledger alone.

Step	Definition	Failure consequence
Target membership	commit's target_id must be in the active pool	zero; logged as leak or fault
Reveal integrity	SHA-256(payload, salt, epoch, target) matches commitment	zero on mismatch or missed reveal
Validity gate	best-candidate TM ≥ 0.35 (delta_tm)	zero below gate
Duplicate gate	3-mer Jaccard vs earlier accepted commits < 0.85 (theta_dup), first commit kept	later duplicates zeroed
Composite	$0.7 \times \text{TM} + 0.3 \times \text{expression term}$	score in $[0, 1]$
Diversity bonus	$+ 0.1 \times \text{mean pairwise } (1 - \text{Jaccard}) \text{ over accepted set}$	re-normalized to $[0, 1]$
Delayed reveal	score sealed until epoch + B (B = 4)	weights consume only revealed scores

Table 2 - The scoring stack

Threat model and measured outcomes

Attack	Mitigation	Simulated result	Verdict
Sybil clone (same generator, same seed, colliding slot)	theta_dup = 0.85 k-mer gate, first commit kept	TM 0.76 but 98-99% zeroed; weight share 0.000	defense holds
Oracle grinding	delayed score reveals, B = 4	no score public before its weight is spent	defense holds
Target leakage	pool membership check on every commit	40 of 40 leaker commits rejected	defense holds
Pool overfitting	half-pool rotation every 16 epochs	rotation executed at epoch 16 in-sim	defense holds
Free-riding (commit, no reveal)	REVEAL hash check	missed reveals zeroed	defense holds
Low-effort spam	delta_tm = 0.35 gate	45 gate failures zeroed in 20 epochs	defense holds

Table 3 - Attacks, mitigations, and 20-epoch simulation results

Parameter sensitivity was kept deliberately boring: thresholds sit far from measured distributions rather than at knife edges. Honest same-target pairs peak below 0.7 Jaccard in the simulation while clones sit at 1.0, so the 0.85 threshold has wide margin on both sides; the 0.35 gate clears 99 percent of honest GA submissions while cutting the bulk of stub-miner failures; and the $B = 4$ delay is long relative to the one-epoch usefulness of a probed score.

8. Implementation Evidence: Simulation Results

The adversarial harness runs the full mechanism over 20 epochs with 32 miners - 12 coupled-GA honest miners, 10 honest stub miners, 6 Sybil clones, 2 lazy miners, 2 leakers - against the 32-target pool, with one chaperone scoring and $B = 4$ delayed weight application. Clones are built the way a real Sybil operator would build them: identical generator and seed as a victim, with the hotkey registered until its slot collides with the victim's target. Everything below is generated by two commands in the repository (pytest and run_simulation) from a fixed seed.

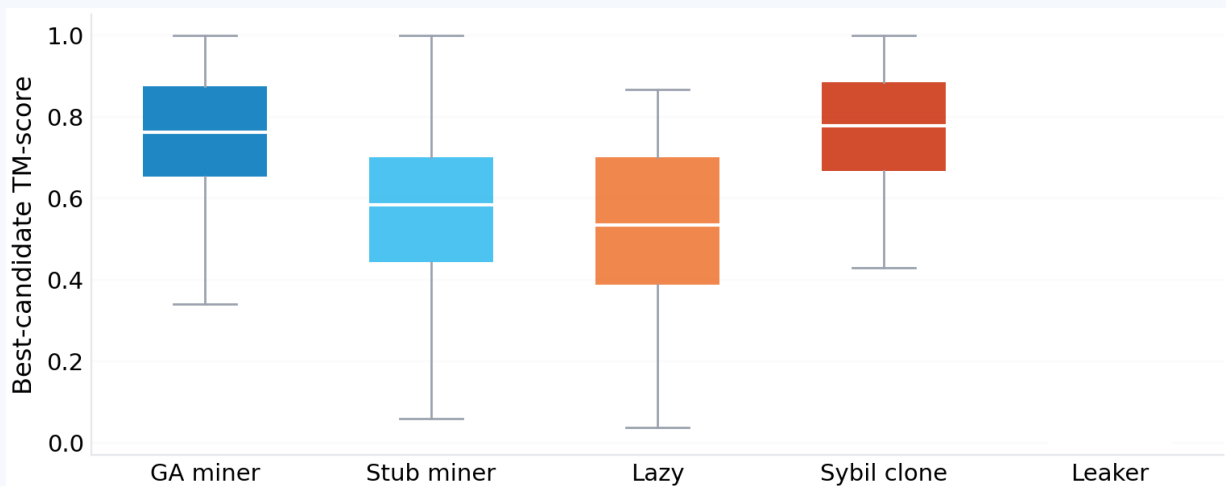


Figure 3 - Best-candidate TM-score by strategy. Clones match victim quality: TM alone cannot catch them.

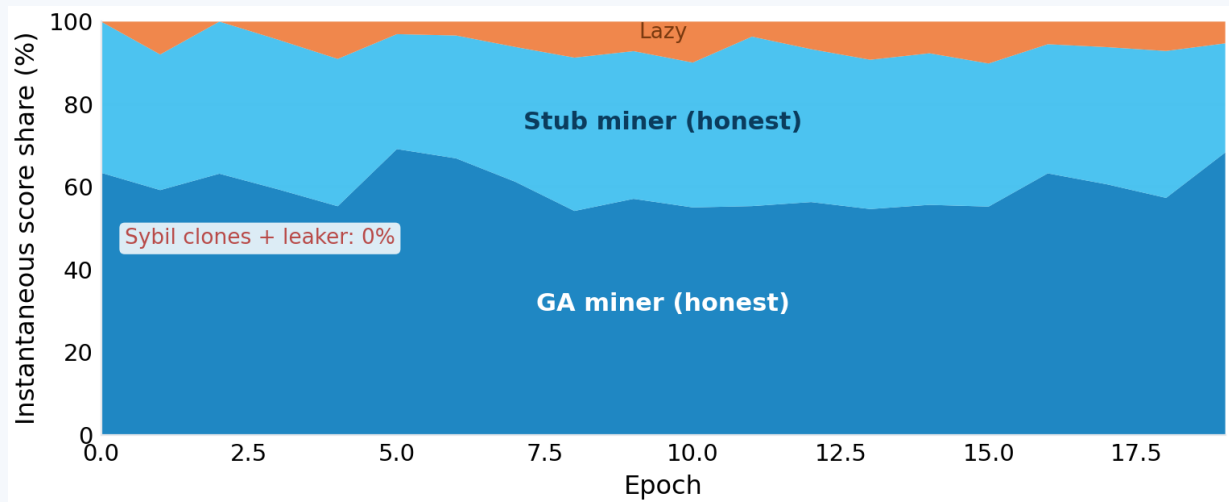


Figure 4 - Instantaneous score share by strategy. Reward concentrates on honest quality; clones are invisible.

Quality and reward separate exactly as designed. GA miners hold 12 of 32 slots (37.5 percent of the population) and take roughly 47 percent of network weight; honest stub miners hold 31 percent of slots and earn 28 percent; clones - folding as well as their victims - earn nothing at all because the duplicate gate fires on their identical payloads before diversity or composite scores are even computed.

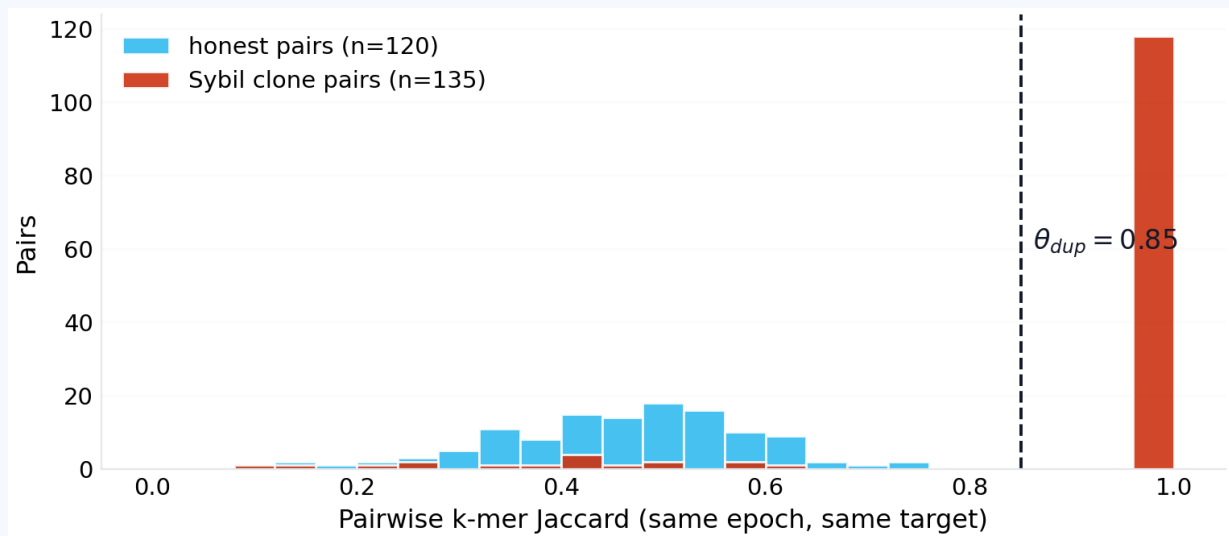


Figure 5 - Pairwise 3-mer Jaccard among accepted or duplicate-flagged miners sharing a target. The θ_{dup} line sits in a wide empty gap.

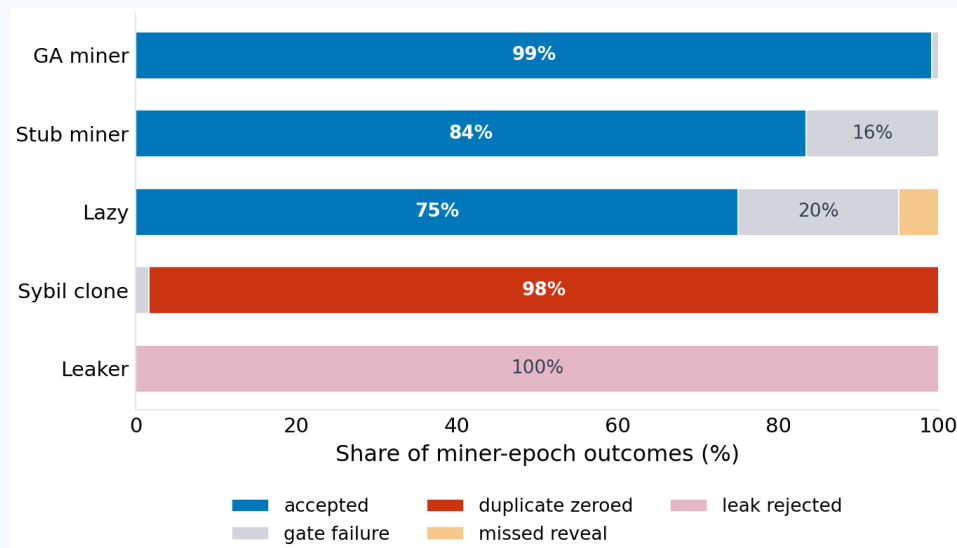


Figure 6 - Outcome decomposition per strategy over 600 miner-epochs. Every attack path earns zero.

The duplicate histogram is the single most important picture in this proposal. Honest miners sharing a target produce sequences whose k-mer similarity clusters well below 0.7; Sybil clones and their victims sit at exactly 1.0. A threshold of 0.85 therefore does not separate two intertwined populations - it separates two populations that never touch, which is the robust regime for a security parameter. Combined with first-commit-kept ordering, the attack economics are final: cloning costs the operator hardware and registration, and returns zero emission, every epoch.

9. Expected Users and Ecosystem Value

The commodity has an existing industry on the demand side. mRNA therapeutic and vaccine developers need sequences that fold correctly and degrade slowly; the OpenVaccine benchmark this subnet's forward model is trained on exists precisely because that pair of properties is the binding constraint. Synthetic biology teams designing riboswitches, aptamers and ribozymes need the same operation at smaller scale. These users consume the network through standard synapses - submit a target structure, receive ranked candidate sequences with oracle scores - or through periodic benchmark reports the network publishes from its own evaluation runs.

For the Bittensor ecosystem the subnet adds three things. First, a clean verification story: folding oracles are deterministic and reproducible, which makes validator consensus auditable in a way subjective commodities are not. Second, a measurable capability curve: because targets rotate from published benchmarks, the network's structural scores over time are directly comparable against academic baselines - the subnet cannot quietly regress. Third, a bridge to computational biology talent: the generator interface admits everything from Kaggle-grade models to state-of-the-art neural inverse folders, so competing on this subnet is meaningful to researchers who have never touched a blockchain. The selected-builder opportunities attached to this hackathon - incubation and accelerator interviews - are the right next step for exactly this reason.

10. Roadmap to Testnet and Mainnet

The project is sequenced so that each phase ends with something that runs. Phase 1, the pre-pipeline of generators, metrics and data loaders, was completed in the companion preprint. Phase 2 - the mechanism layer submitted here - is finished and tested: commit-reveal ledger, duplicate gate, diversity bonus, delayed score reveals, pool rotation, both neurons in mock mode, the adversarial harness, and 58 green tests. Phase 3 is the hackathon final on October 19: three validators and ten miners on Bittensor testnet, a demo video recorded from the live network, and ViennaRNA replacing the stub oracle. Phase 4 is production: RhoFold+ 3D refolding, the trained stability forward model inside the miner ranking path, an RNASolo-sourced target pool, and GPU-backed miners.

Phase	Scope	Status	Evidence
1. Pipeline	generators, metrics (TM / RMSD / base-pair), data loaders	done (preprint)	companion paper + repo modules
2. Mechanism	commit-reveal, duplicate gate, diversity, delayed reveals, rotation, harness, tests	done - this submission	58 tests, 20-epoch run, charts
3. Testnet	3 validators / 10 miners on testnet, ViennaRNA oracle, demo video	Oct 19 target	live network demo
4. Production	RhoFold+ 3D oracle, XGBoost miner ranking, RNASolo pool, GPU miners	post-hackathon	incubation plan

Table 4 - Phase status and deliverables

The seven-day operational plan through testnet is committed in the repository (docs/SEVEN_DAY_PLAN.md): testnet skeleton on day 2, ViennaRNA oracle swap on day 3, forward-model ranking on day 4, the real RNASolo target pool on day 5, and demo recording with buffer on days 6 and 7. Each day ends with a runnable artifact, so slippage costs a day, never the submission.

References

The mechanism and miner technology in this proposal come from two companion preprints by the team, supplied as hackathon materials; the remaining citations anchor the oracle, dataset and benchmark choices.

- **The Ribosome Network: A Decentralized RNA Inverse-Folding Subnet as a Synthetic Ribosome.** Team preprint, 2026. Mechanism source: roles, phase schedule, $\theta_{\text{dup}} = 0.85$, $w_{\text{div}} = 0.1$, $N_{\text{pool}} = 32$, $T_{\text{rot}} = 16$, $B = 4$.
- **Stability-Aware mRNA Inverse Design via Coupled Forward-Model-Guided Search.** Team preprint, 2026. Miner source: coupled GA, beta ablation (Table 5), XGBoost forward model MCRMSE 0.356 plus or minus 0.004.

- **Wayment-Steele, H. et al.** Probing RNA dynamics via live NMR and self-supervised machine learning. OpenVaccine dataset, Stanford COVID-19 mRNA Vaccine Degradation Prediction, 2020.
- **Das, R. et al.** RNA structure prediction: critical assessment (benchmark set used for the production target pool), 2010; RNASolo structure repository.
- **Shen, T. et al.** RhoFold+: RNA 3D structure prediction. 2022. Phase-4 oracle; **Lorenz, R. et al.** ViennaRNA Package 2.0. Algorithms for Molecular Biology, 2011. Secondary-structure oracles.

Appendix A - Mechanism hyperparameters

Parameter	Value	Meaning
EPOCH_LENGTH_SEC	900	one round, 15 minutes
PHASE_DURATIONS	3/6/2/2/2 min	COMMIT / EVALUATE / SET_WEIGHTS / REVEAL / ROTATE
N_POOL / T_ROT	32 / 16	active targets; half-pool rotation period (epochs)
K_CANDIDATES	4	sequences per miner per epoch (paper default 8)
DELTA_TM	0.35	validity gate on structural score
W_STRUCT / W_EXP	0.7 / 0.3	composite score weights
SCORE_REVEAL_DELAY_B	4	epochs a score stays sealed
THETA_DUP / KMER_K	0.85 / 3	duplicate Jaccard threshold; shingle width
W_DIV	0.1	diversity bonus weight
VALIDATOR_QUORUM / TRIM	5 / 1	quorum size; trimmed-mean trim
GA_BETA (sim)	2.0	coupled-objective weight; production guidance 0.1-0.5
GA_POP / GENS (sim)	12 / 4	sim budget; paper scale 200 x 60

Table 5 - Complete mechanism constants (ribosome/constants.py)